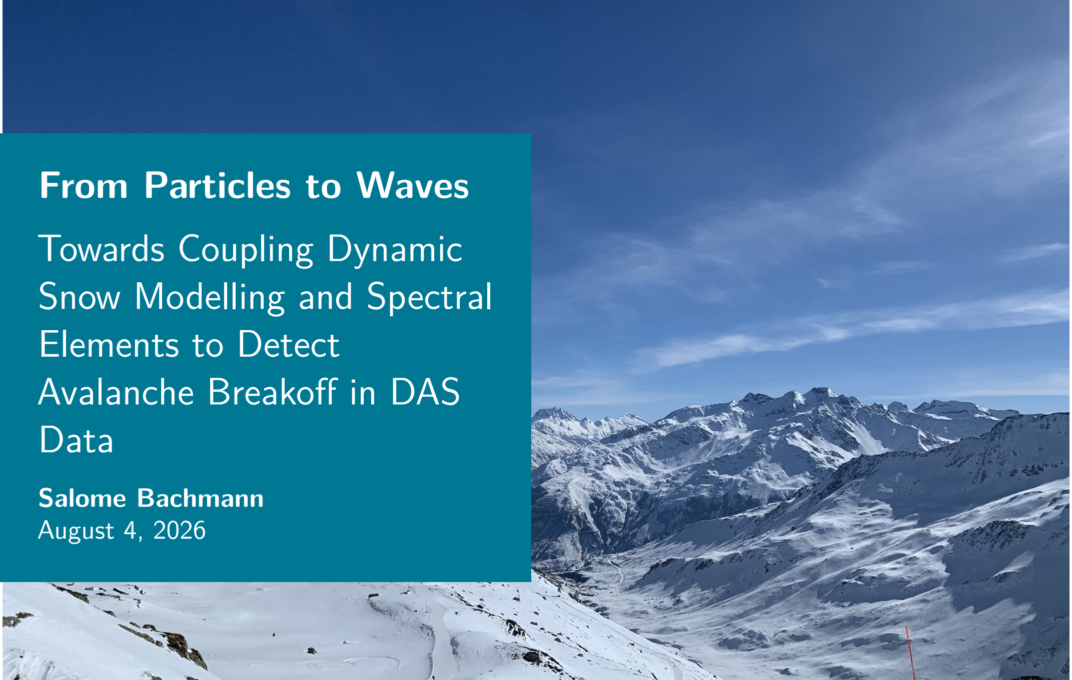

From Particles to Waves

Towards Coupling Dynamic Snow Modelling and Spectral Elements to Detect Avalanche Breakoff in DAS Data

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Overview

- Created 3 different models and 2 scenarios for crack propagation in snow using SALVUS
- Two pure spectral element (SE) models, one combined with data from material point method (MPM) simulations
- **Goal**: Couple dynamic snow modelling with SE to model crack propagation in snow

Why model crack propagation in snow? - Relevance

- 22 fatalities by avalanche in Switzerland every year (SLF, 2025)
- Avalanches are danger for human life but also infrastructure (Schweizer et al., 2003)
- Understanding processes associated with avalanches important
- Avalanche release because of loading, leading to crack propagation in a weal layer (Gaume et al., 2017)

Relevance

- Crack propagation at different speeds: sub-Rayleigh (below vs of material) and Supershear (above vs) (Trottet et al., 2022), transition between two regimes possible (Bergfeld et al., 2025)
- Fast crack propagation $\rightarrow$critical crack length for release (Gaume et al., 2015) is reached faster
- Distributed Acoustic Sensing (DAS) already used to monitor avalanches in Switzerland (Edme et al., 2023; Paitz et al., 2023) and Norway (Turquet et al., 2024)
- No knowledge if crack propagation associated with breakoff can be seen in DAS data

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